Koen Essers • Master Internship • October 14, 2026

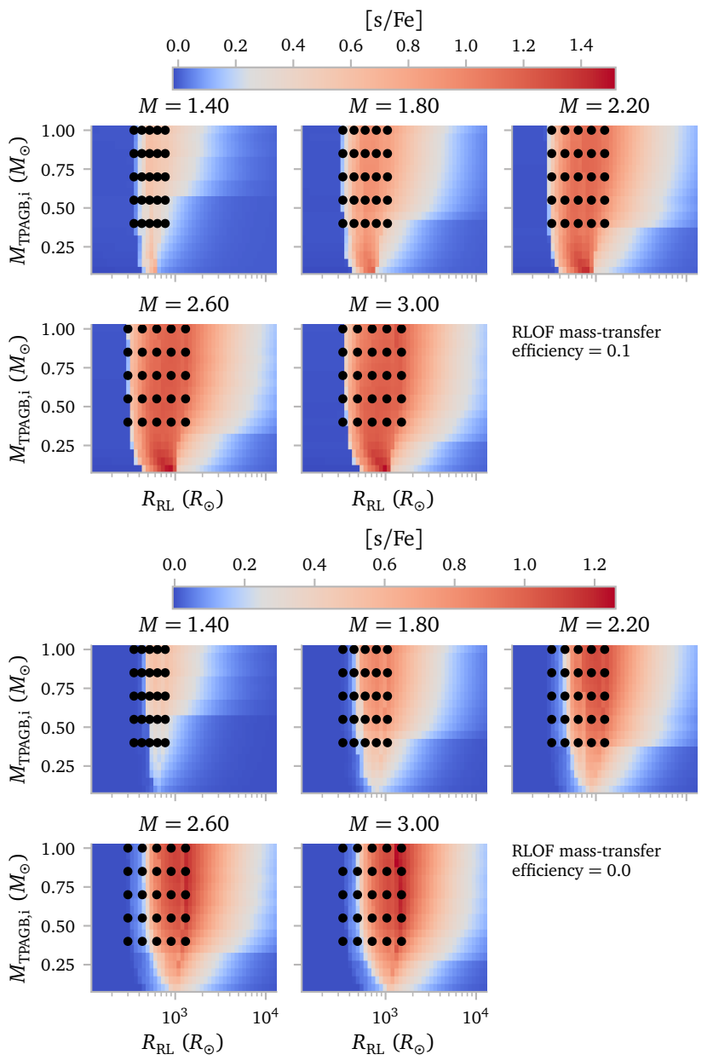
Notes Week 32

a. Grid

Based on the simple binary evolution grids I showed last week, I have decided to simulate a high dimensional grid that looks at a broad mass range, broad initial mass-ratio range, and initial Roche lobe radii such that I simulate pretty much the complete RLOF regime with some dredge up.

I do this for two RLOF accretion efficiencies, each with three different ratios for isotropic re-emission and circumbinary mass loss. The ratios are (CCT, isotropic re-emission) = (0.1, 0.9) ,(0.2, 0.8) ,(0.3, 0.7). The accretion efficiencies are 0 and 0.1.

The plot below shows the complete grid.



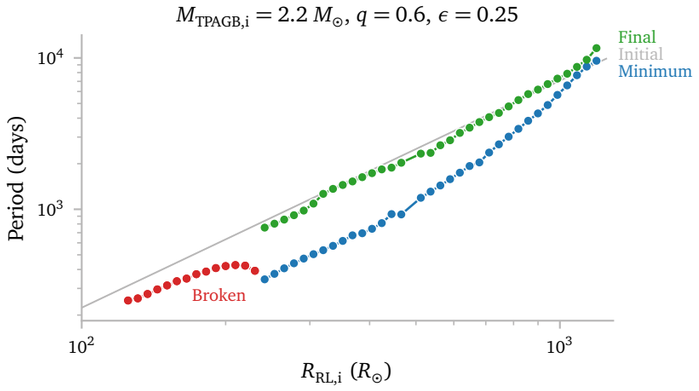
Notice in naive simple binary case, the different ratios between isotropic re-emission and mass loss to a circumbinary torus would result in the same accretion and thus in the same abundances. Therefore, each dot is really 3 dots stacked on top of each other.

In the detailed MESA simulation, there will most likely be small differences between the final abundances for these systems. However, the biggest differences between these will be in the final orbital properties of these systems. More mass loss to a circumbinary torus will generally lead to much tighter final binary systems.

The panels with $\varepsilon_{\mathrm{RLOF}} = 0.1$ and $\varepsilon_{\mathrm{RLOF}} = 0.0$ at first sight might look very similar. This is because in both these cases, *most* of the mass is transferred via winds, which are not affected by the efficiency parameter. However, in the regime where winds are not very strong, differences between the two settings can be seen. This is on the left side of the plot. Here we see that the systems with *some* RLOF accretion are generally able to create barium stars with slightly smaller initial Roche lobe radii.

b. Roche lobe radius Grid

Below I will compare some of the orbital properties of the broad Roche lobe radius grid that I simulated a couple of weeks ago.

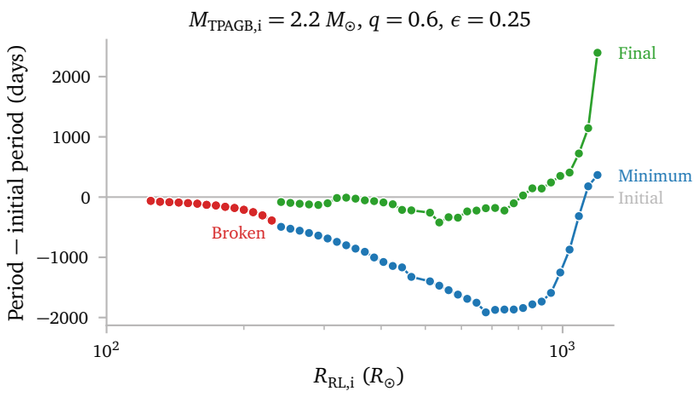


This plot shows the period of the various models. The gray line shows the initial period of the models as a function of the initial Roche lobe radius. This is simply a power law based on the Eggleton (1983) Roche lobe radius approximation and Kepler's Third Law.

I also plot the final period, which is in some cases *smaller* than the initial period, and in some cases, especially towards large initial Roche lobe radii, the final period is *larger* than the initial period.

The minimum period get reached during RLOF and is always smaller than the final Roche lobe radius.

It is more obvious if we plot this compared to the initial period.

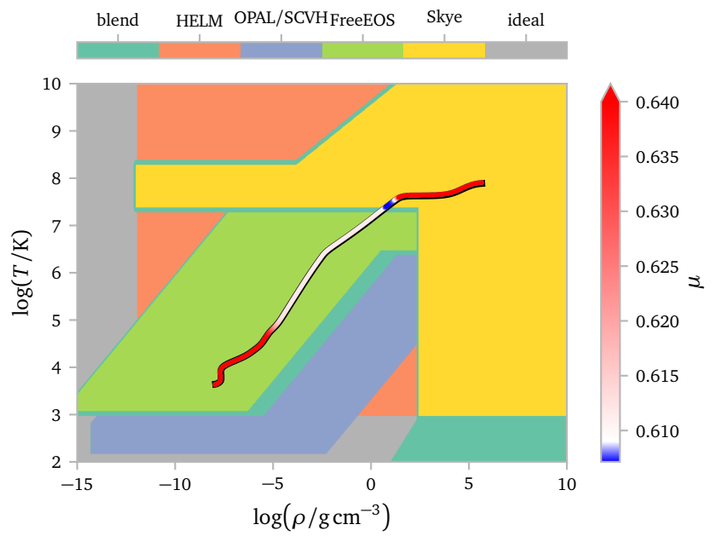


The differences with respect to the initial period increases with the Roche lobe radius. This is in part due to the tides in combination with the winds, and due to the RLOF, which seems to decrease the periods *more* for the larger masses.

c. Mean Molecular Weight

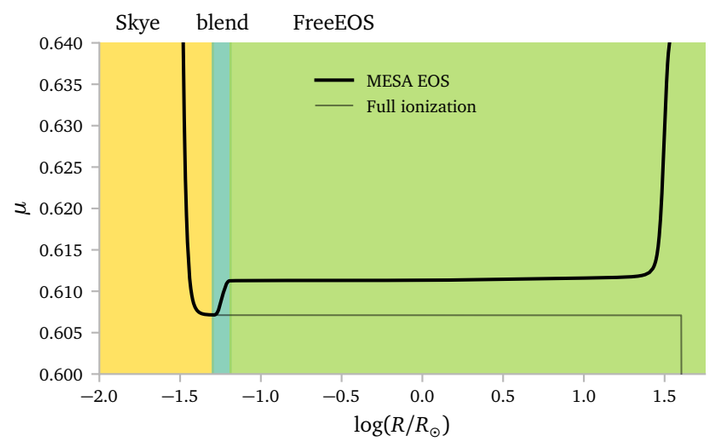
c.1 Giant Branch Dip

Using the `EOS_plotter` routine from *MESA*, it is possible to clearly plot the different EOS tables that MESA uses, and in which regions in the ρ-T diagram. Below, I show this, and I plot the ρ-T profile for an evolved RGB star over these tables.



The RGB profile is colored, where very red regions correspond to regions with relatively large mean molecular weight μ. Blue regions correspond to the lowest mean molecular weights, and white corresponds to an average mean molecular weight of the envelope. I think this figure quite clearly shows why we see a pit in the mean molecular weight. The core boundary happens inside the Skye EoS table (Jermyn et al., 2021). This table *assumes* complete ionization. Once the profile crosses over to the boundary region, we see that the mean molecular weight increases until it crosses the FreeEOS regime, which does *not assume* complete ionization.

Now that the regions of the different equations of state tables are clear, and we know that our giant branch models cross into the Skye EOS just before the core, we can again make a plot of the μ as a function of depth in the star. For the sake of clarity, I chose $\log R$ for the horizontal axis, but obviously one could chose various different quantities such as internal mass, temperature or pressure.

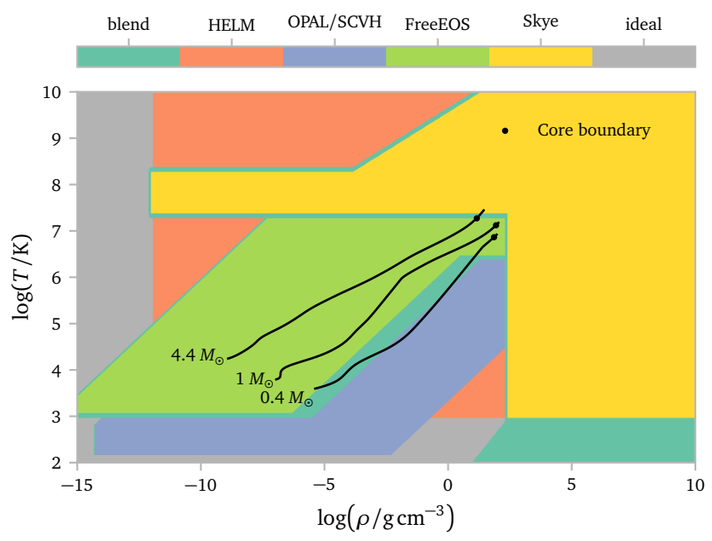


This plot *clearly* shows that the pit in the μ profile is not real. The Skye EOS (Jermyn et al., 2021) *assumes* complete ionization, which, according to the FreeEOS, is clearly not true up until the deepest parts of its influence. The cyan blending region just ensures the two different tables do not produce discontinuous profiles.

We also clearly see that my naive complete ionization prescription to compute the mean molecular weight perfectly matches the Skye (Jermyn et al., 2021) EOS, and starts to diverge in the blending region, just as I would expect based on the fact that Skye assumes full ionization and FreeEOS does not.

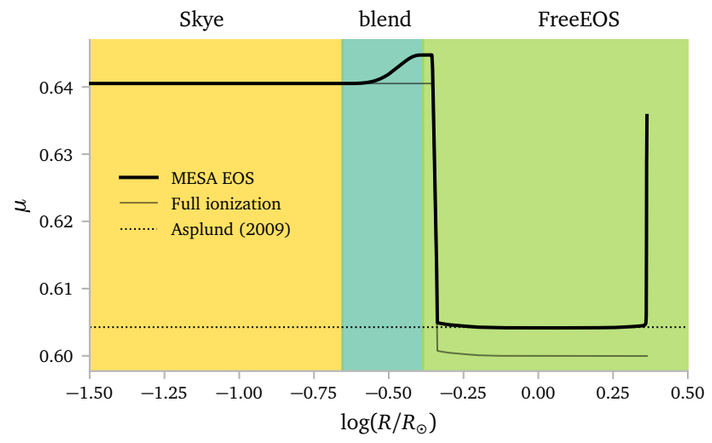
c.2 Main Sequence

Now, let's look at the main sequence. Maybe here we see a similar crossing between EOS regimes.



I once again plot the ρ-T plane and the regions of the different EOS tables. I then plot 3 different MS models with different masses and $X(\mathrm{H})_{\mathrm{center}} = 0.65$. I also indicate where the hydrogen burning core boundary is with a black dot.

The plot clearly shows that the heaviest model here crosses into the Skye EOS (Jermyn et al., 2021) in its core. The entirety of the 1 $M_\odot$ model uses the FreeEOS table and the 0.4 $M_\odot$ crosses into the OPAL/SCVH region (Rogers & Nayfonov, 2002).

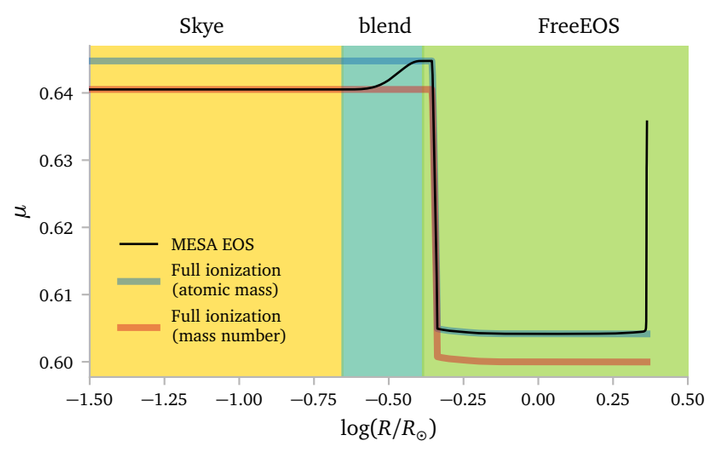


Again, we see that the mean molecular weight of the Skye EOS agrees perfectly with my fully ionized computation. As soon as we enter the blended region, the mean molecular weight starts to deviate.

The weird thing is that the Asplund et al. (2009) mean molecular weight agrees with the FreeEOS nearly perfectly.

It took me an embarrassingly long time, but I *think* I have finally been able to figure out what causes the differences in μ between the Skye EOS and FreeEOS. The differences can be explained by the masses of the elements that are used in the computation of μ. My naive approach for complete ionization for the *MESA* abundances was to simply take integer mass numbers equal to the number of protons plus neutrons. *However*, for the Asplund et al. (2009) mean molecular weight computations, I use the Standard atomic weight. This is a bit different.

It seems like Skye (Jermyn et al., 2021) uses the mass number, while the FreeEOS uses the atomic mass. In fact, when I compute the *MESA* mean molecular weight using both methods, we can clearly see that the computed mean molecular weights match depending on which mass is used.



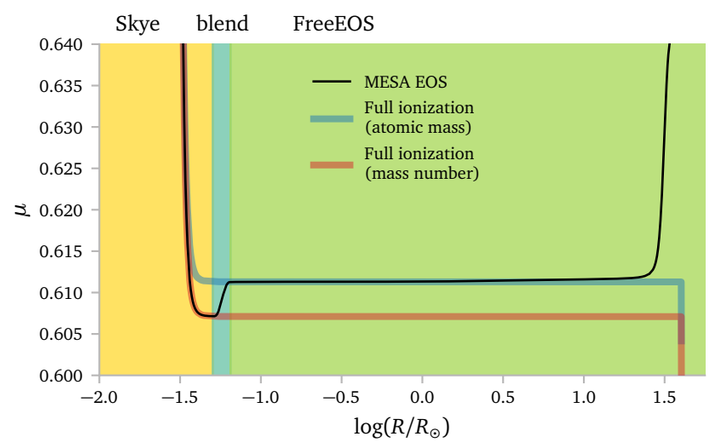
Clearly, the "mass number" mean molecular weights match the Skye (Jermyn et al., 2021) EOS mean molecular weights. The "atomic mass" mean molecular weights match the FreeEOS mean molecular weights.

This is not necessarily a problem. For all main sequence models, the blending region, if it even exists, is *inside* the core. This means that for the entire mixing region, the FreeEOS mean molecular weight is used. I use this same "atomic mass" prescription to compute μ for the accreted material, so to compute the mixing mass, we can simply use the full ionization with atomic masses. There are then no inconsistencies.

I have now implemented the fully ionized mean molecular weight computation into the accretor code. Since it agrees so well with the normal *MESA* EOS, there are no noticeable differences in the abundances or mixing masses.

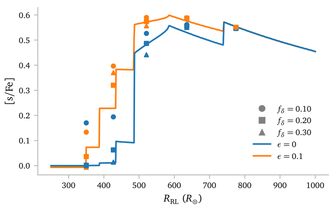
c.3 Giant Branch Dip 2

I also plot the two different computation methods for the late RGB model.



Again, clearly the "mass number" mean molecular weights match the Skye (Jermyn et al., 2021) EOS mean molecular weights. The "atomic mass" mean molecular weights match the FreeEOS mean molecular weights.

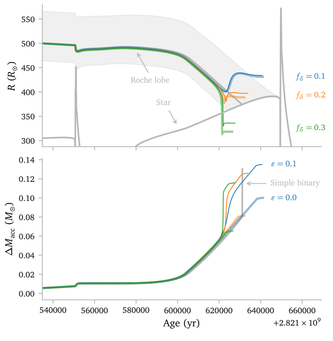
d. 1.4 Solar mass grid



Here, I show the final [s/Fe] ratios for $q = 0.85$, $M_{\mathrm{TPAGB}} = 1.4\,M_\odot$ detailed *MESA* binary simulations and the simple binary simulation.

I think the *MESA* models and the simple binary models correspond quite well, but there are definitely differences. Mainly, the detailed MESA simulations seem to predict slightly larger observed abundances in many cases. For one model however, the abundances are smaller. Then, for the wind only regime, the *MESA* models agree very well with the simple binary models.

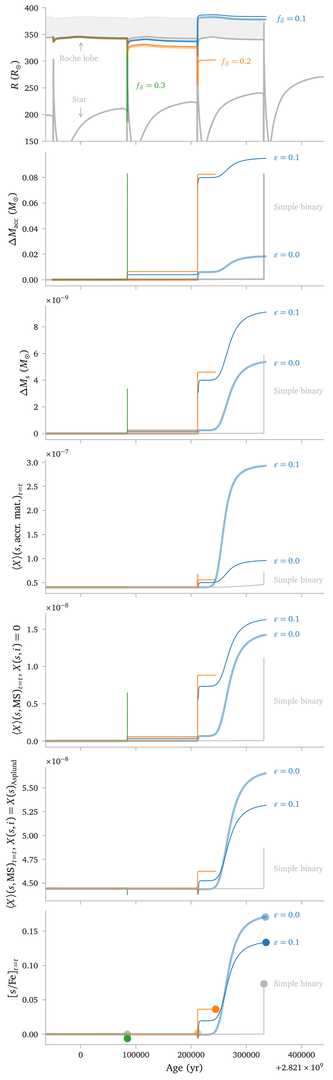
Let me first focus on the set of models with the third Roche lobe radius. These are interesting because they interact during the phase of strong winds. We can recognise this in the figure above by the clear upwards diagonal in the $\varepsilon = 0$ models. The downwards sloping diagonal line that follows it contains models that interact during the last thermal pulse.



Here, I plot the Roche lobe radius and the star radius as a function of time for the models with an initial Roche lobe radius of approximately $R_{\mathrm{RL}} = 520\,R_\odot$. The most obvious differences between the models is caused by the differences in f_δ, which is to be expected. We see that more mass loss to a CCT leads to smaller final periods. A second, initially less obvious effect is that these models interact *earlier*. This effects how much mass is lost through *winds*.

If we look at the total amount of accreted mass, we can clearly explain the differences we see between the simple binary and the detailed *MESA* simulations. For the $\varepsilon = 0$ case, the earlier RLOF interaction causes less mass to be transferred via winds for the $f_\delta = 0.3$ model. For the $f_\delta = 0.1$ model, there are still significant winds during and after RLOF, which invalidates the assumption of the simple binary that all mass is lost through RLOF once this starts. For the $\varepsilon = 0.1$ models, this effect causes the differences.

Now, let me look at the first Roche lobe radius, where we see some peculiar differences, especially for the $\varepsilon = 0, f_\delta = 0.1$ model.



Apparently, these different mass transfer mechanisms cause these models to interact during different thermal pulses. This makes it slightly confusing to interpret the differences between the simple binary and the detailed *MESA* models. To start, we see that the $f_\delta = 0.3$ models interact first. This means these models will transfer mass, independent of how much, with the *lowest* enrichment. In fact, the envelope is barely enriched at all during this thermal pulse and therefore, both of these models do not show any enhanced [s/Fe]. Then, the orange $f_\delta = 0.2$, models. These interact one pulse later, but still earlier than the simple binary. This means that these models will have lower, but still slightly enhanced envelope abundances, which explains why the model with $\varepsilon = 0.1$ predicts a slightly enhanced [s/Fe], but still lower than the simple binary. Lastly, we have the $f_\delta = 0.1$ models. These undergo RLOF during the same thermal pulse, *but* they do not expel their entire envelope during RLOF, unlike what we assume for the simple binary. Because the envelope becomes enriched due to dredge up *after* the RLOF episode, wind mass losses after this episode are especially important. Because the $\varepsilon = 0.0$ model transfers no not very enriched material during RLOF, and about the same amount of enhanced material with winds, it predicts the highest [s/Fe] value.

Let's look at this a bit more carefully. The second and third panel show the amount of accreted mass, and the amount of accreted s-process mass. Clearly these are both larger for the $\varepsilon = 0.1$ model, which makes sense, because it is able to transfer mass during RLOF.

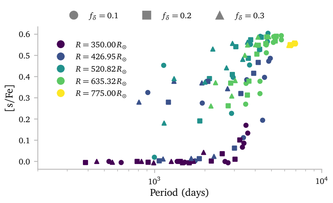
The subplot on the left shows the averaged abundance of the material that is accreted. It is important to realise that this is *not* the instantaneous abundance of the material that is accreted at time t. Because a strong third dredge up episode enhances the envelope of the TPAGB star after RLOF in the blue models, the abundances of s-process elements are much larger after this dredge up. This means the averaged abundance of the accreted material is much larger for the $\varepsilon = 0$ model than for the $\varepsilon = 0.1$ model.

We can then accrete this material onto the accretor. In this case I use a constant mixing mass that is determined based on the final value, so *strictly speaking*, only the last value is accurate. The subplot to the left and the one above it show the s-process abundances based on this assumption. The first being *no* initial *s*-process elements in the accretor, and the second using more realistic initial s-process abundances based on Asplund et al. (2009). Without initial abundances, the $\varepsilon = 0.1$ model generates the largest accretor abundance, simply because it accretes more s-process elements. However, because the bulk of this accreted material has abundances that are very close to the initial value, these really do not contribute much if we consider an accretor with initial s-process abundances.

Finally, if we use the correct mass-fractions, we can compute [s/Fe] for the models, and we obtain the final values.

I feel like, with this, I can explain the differences between the simple binary and the detailed *MESA* models. They originate from different RLOF moments and the assumption of complete RLOF mass transfer.

Now, let's look at some more general properties. To start, let me plot the [s/Fe] abundances as a function of the final period of the binary.



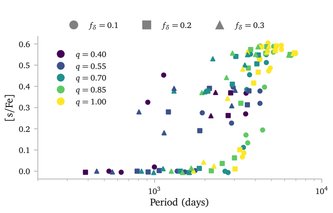
With a broader period range than before, we can actually see some structure. Clearly, for small initial Roche lobe radii, non of the other parameters actually matter for the [s/Fe] since there is no dredge up in the TPAGB at the point of RLOF.

Then there is a clear clustering of many models with large period of several thousands of days and [s/Fe] abundances around 0.5 − 0.6. Lastly, there are some systems with slightly lower [s/Fe], but *much* shorter periods. These are not exclusively systems with $f_\delta = 0.3$, like I would expect.

I think these systems are a combination of two effects. First, we have that the $\delta = 0.3$ models have much stronger orbital shrinkage, which moves them to lower periods from the right. So the clustered systems on the upper right of the plot migrate to *lower periods* when they have a higher δ.

Then secondly, we have the effect where systems with lower q respond more intensely to mass loss, and therefore obtain much shorter periods and in many cases also lower abundances.

We can confirm this by using q as the color instead of the initial Roche lobe radius.

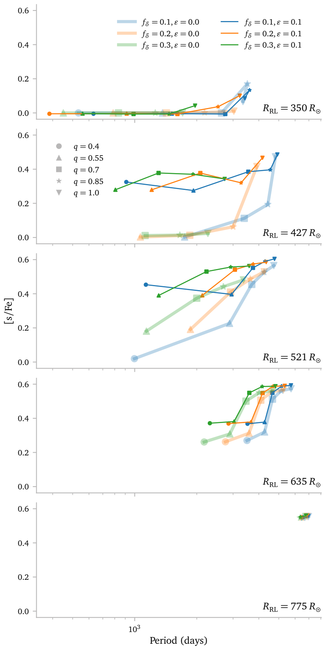


We quite clearly see that most of the sparse, floating points are either

1. systems with small q,
2. systems with large δ,

or both.

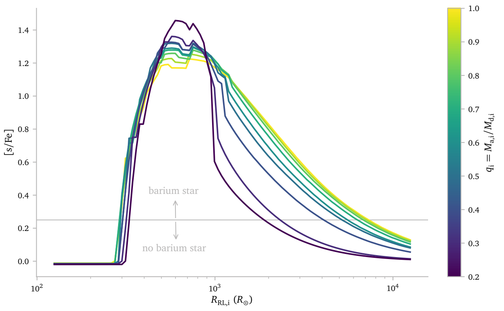
This effect becomes even clearer when we plot only a single Roche lobe radius at a time.



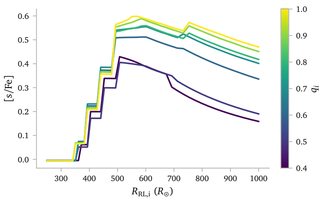
Clearly, the Roche lobe radius has the largest effect on where in the this diagram the points end up, *but* the different q values can have a significant role in shifting the models, especially to lower periods.

This this point, the models do not experience RLOF and therefore the different mass loss mechanisms during RLOF do not matter anymore. Only q is relevant, but only the models with the two *largest* q values were actually simulated in this case. .

Intuitively, based on the figure I show below that I made last week, you might think that the lower q values cause lower accretor masses, and therefore higher abundances during RLOF. The models below are for a TPAGB star with mass 2.2 $M_\odot$ and $\varepsilon_{\rm RLOF} = 0.25$.

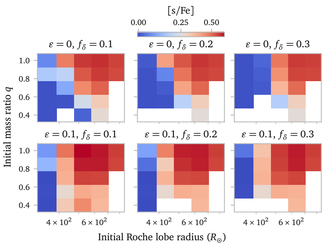


However, the story is a bit more nuances than that. Obviously the accretion efficiency of RLOF plays a role, because if we would have $\varepsilon = 0$, then only wind matters and wind accretion is generally less effective for lower q. Then we also have the effect that models with lower q interact earlier, which, in combination with dredge up can significantly lower the abundances of the accreted material.

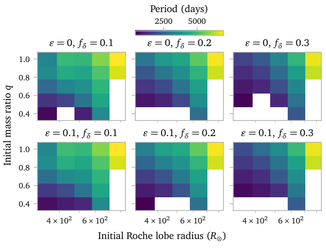


Below we see that, if we take into account these nuances, it is very much possible to quite significantly lower s-process enhancements for models with lower q in the RLOF dominated regime. These are models for 1.4 $M_\odot$ and $\varepsilon_{\rm RLOF} = 0.1$.

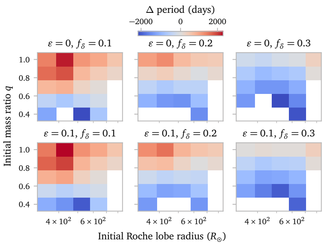
We can of course make some 2d plots.



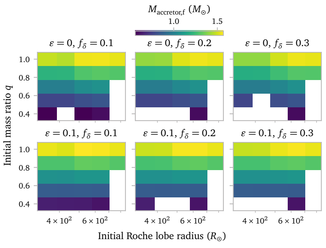
Here I compare the [s/Fe] abundances. The un-enriched regime on the left is clearly larger for the $\varepsilon = 0$ models. This is what we expect. We then also see that the actual s-process abundances in the region where both become enriched are not very different. It seems like even this low mass star can enrich the accretor enough to create a barium star according to the definition of [s/Fe] > 0.25. There are three models missing on the bottom right of each panel. These models have not been simulated because they would miss RLOF by too much. The other empty spots are due to inspirals. None of these models crashed.



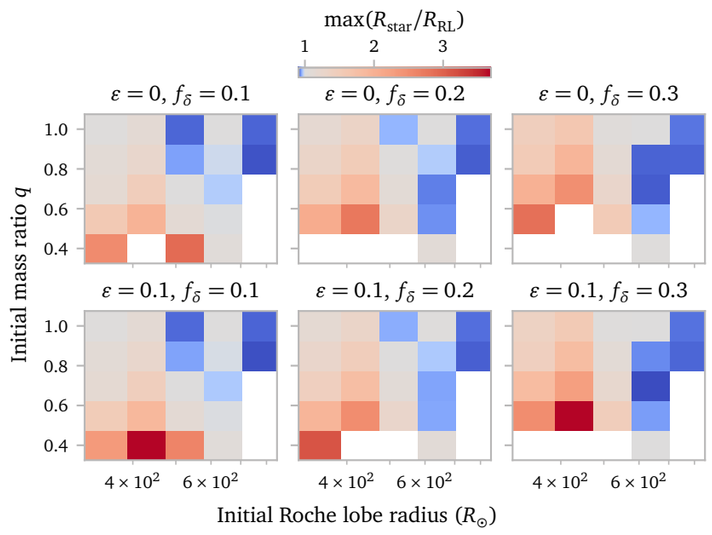
The periods are distributed like we have seen many times before. No surprises here.



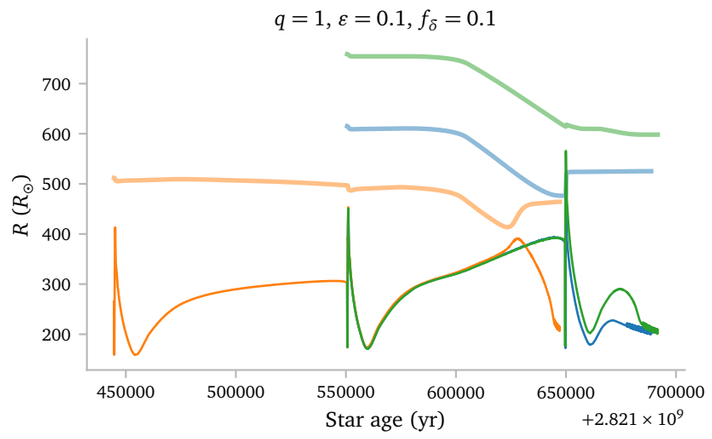
If we look at the difference between the initial period and final period, we clearly see that this is strongly affected by the mass ratio and f_δ. It is difficult to say if the initial Roche lobe radius has any noticeable, but we have seen in one of the plots above that it can have an effect.



Here I plot the final mass of the accretor after the full binary evolution. The patterns make a lot of sense, and many of the final masses actually overlap with the masses of observed barium stars, which is nice.

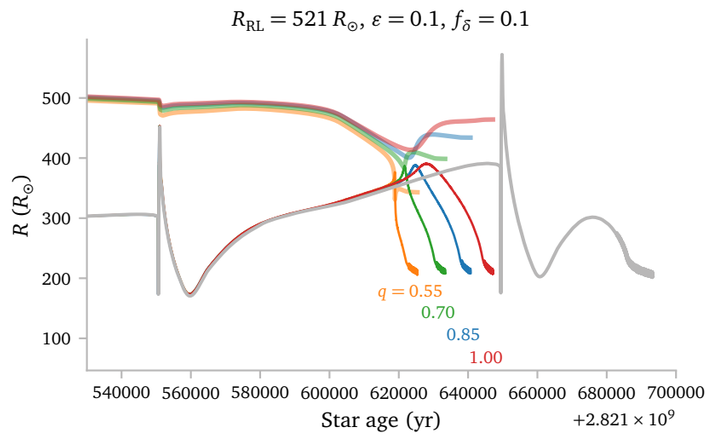


This is very surprising to me. So much so that I actually had to make sure I was really plotting the correct quantity. We see *much* less extreme overfill ratios for these 1.4 $M_\odot$ models than we do for the heavier stars. In some cases, the star actually undergoes RLOF, but it never reaches exceeds its own Roche lobe radius. Let me plot one of these models.



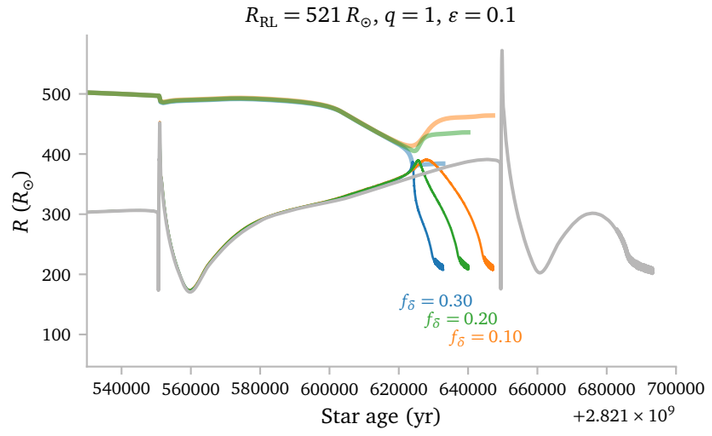
The surprising shape is not surprising anymore if we look at models with equal initial mass ratios, but different starting Roche lobe radii (so rows in the plots above). We see that in the case of equal mass ratios and not very extreme circumbinary mass loss, models that approach RLOF during the interpulse can actually strip their envelope with RLOF without large overfill ratios. As soon as the models reach the last thermal pulse, we get a region again where the models can obtain large overfill ratios due to the thermal pulse. Then beyond a certain initial Roche lobe radius, the binary will not experience RLOF at all and we obviously get small overfill ratios.

Instead of looking at a row, we can also inspect a column. Clearly an interesting column to look at is the one where some models do overfill their Roche lobes, and some do not.



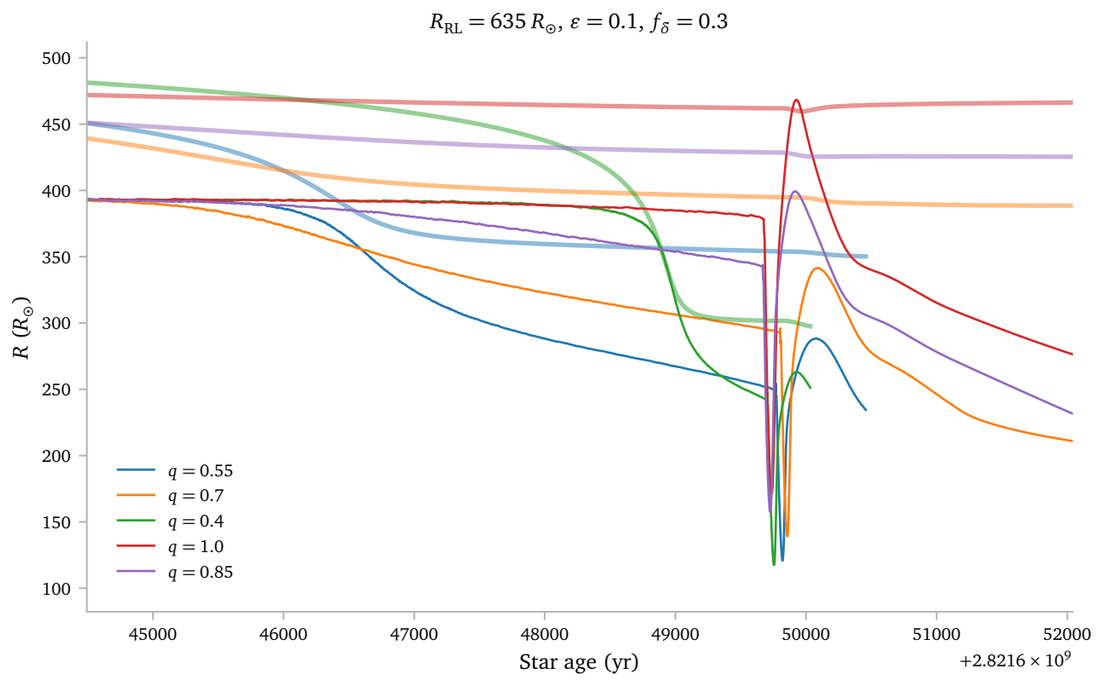
Apparently, q, has quite a large influence on *if* the model reaches significant overfill ratios. This makes sense considering what we have seen much earlier, where the models with the most unequal mass ratios also have the most extreme RLOF episodes, the most instabilities and the possibility to spiral in due to tides.

We can also compare *across* the plots. For example, let me compare the 3 different $f_{\rm delta}$ values.



The effect of this is similar to what we see for different q values prior during RLOF, which is what we expect since *both* of these parameters alter the severity of RLOF. However, an important difference between the two is that the mass ratio *also* affects the winds, while δ does *not*. Therefore, prior to RLOF, the models with different δ will behave more similarly than the models with different q.

Lastly, let me different q values again, but now shifted one column to the right.



This is quite interesting. We see that the $q = 1$ model *does* overfill its Roche lobe, while for example, the other models do not overfill their Roche lobe during the thermal pulse. However, the $q = 0.4$ model slightly overfills its Roche lobe during RLOF slightly before the thermal pulse.

References

Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. (2009) *The Chemical Composition of the Sun*. ARA&A47, 481.

Eggleton, P. P. (1983) *Aproximations to the radii of Roche lobes.*. ApJ268, 368.

Jermyn, A. S., Schwab, J., Bauer, E., Timmes, F. X., & Potekhin, A. Y. (2021) *Skye: A Differentiable Equation of State*. ApJ913, 72.

Rogers, F. J. & Nayfonov, A. (2002) *Updated and Expanded OPAL Equation-of-State Tables: Implications for Helioseismology*. ApJ576, 1064.